

DEPARTMENT OF ELECTRONIC & COMPUTER ENGINEERING > Programs > Undergraduate Programs > **BEng (ELEC)**



BENG IN ELECTRONIC ENGINEERING

Degree Awarded: BEng in Electronic Engineering

JUPAS Code: JS5250 (through admission to the Department of Electronic and Computer Engineering)

Abbreviated Title: BEng (ELEC)

Program Duration: Normally 4 years

Programme Outcomes(PO)

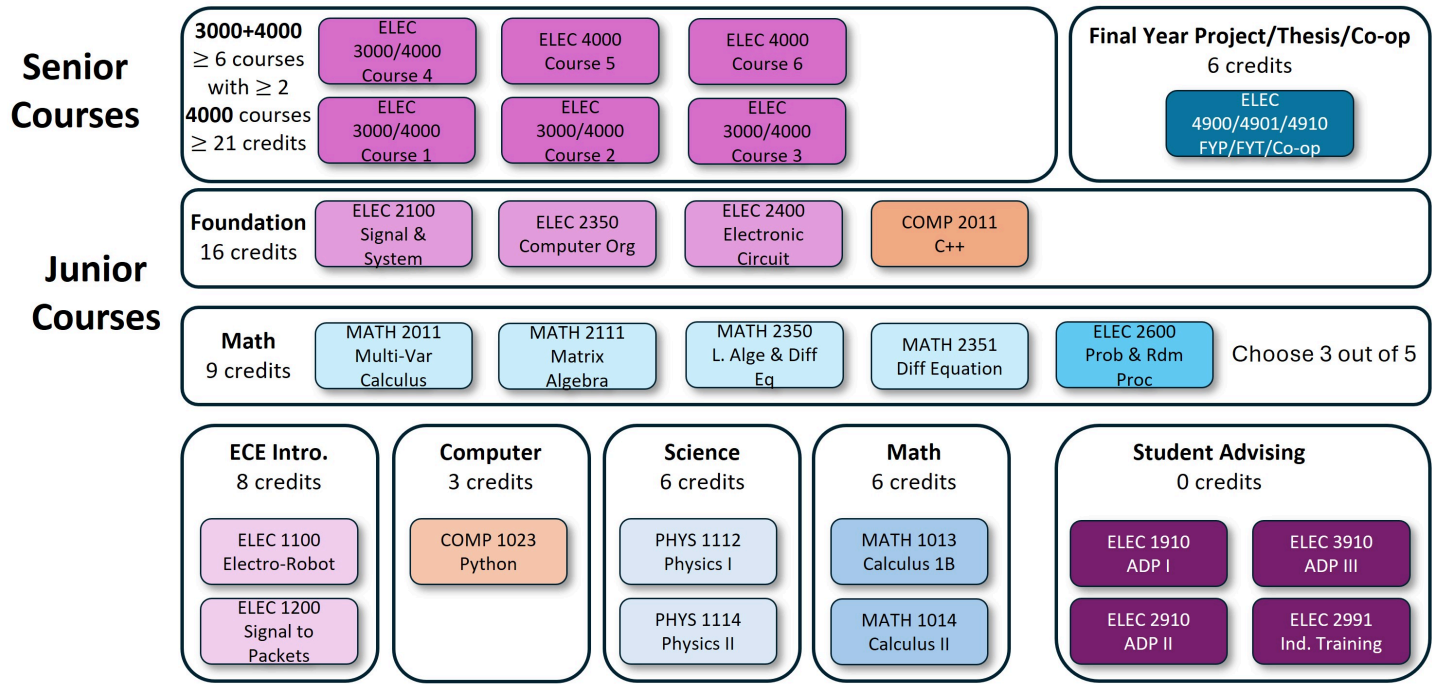
PO1	An ability to apply knowledge of mathematics, science and Electronic and Computer Engineering.
PO2	An ability to design and conduct experiments, as well as to analyze and interpret data.
PO3	An ability to design efficient and economical Electronic and Computer Engineering systems, components or process subject to practical constraints.
PO4	An ability to function in a multi-disciplinary environment through teamwork.
PO5	An ability to identify, formulate and solve Electronic and Computer Engineering problems.
PO6	An ability to understand professional practices and ethical responsibilities.
PO7	An ability to communicate effectively.
PO8	An ability to understand the impact of electronic engineering solutions in a global, economic , and societal context, especially the importance of health, safety, and environmental considerations to both workers and the general public.
PO9	An ability to recognize the need for, and to engage in life-long learning.
PO10	An ability to use current techniques, skills and engineering tools necessary for solving Electronic and Computer Engineering problems.

The Curriculum

The Department of Electronic and Computer Engineering (ECE) has developed a unique and special major program for the 4-year curriculum following Outcome-Based Education (OBE). The curriculum design is based on being rewarding, relevant and fun while allowing for depth, breath and flexibility. We aim to enable students to learn fundamental ECE concepts and have balanced exposure to both engineering design and analysis, and help students develop an engineering problem-solving mindset.

Students will be admitted first to the School of Engineering rather than to individual programs. They will have at least one year to explore various programs before deciding on their major.

ELEC Curriculum (2025/26 or later)



Graduation: 120 Credits

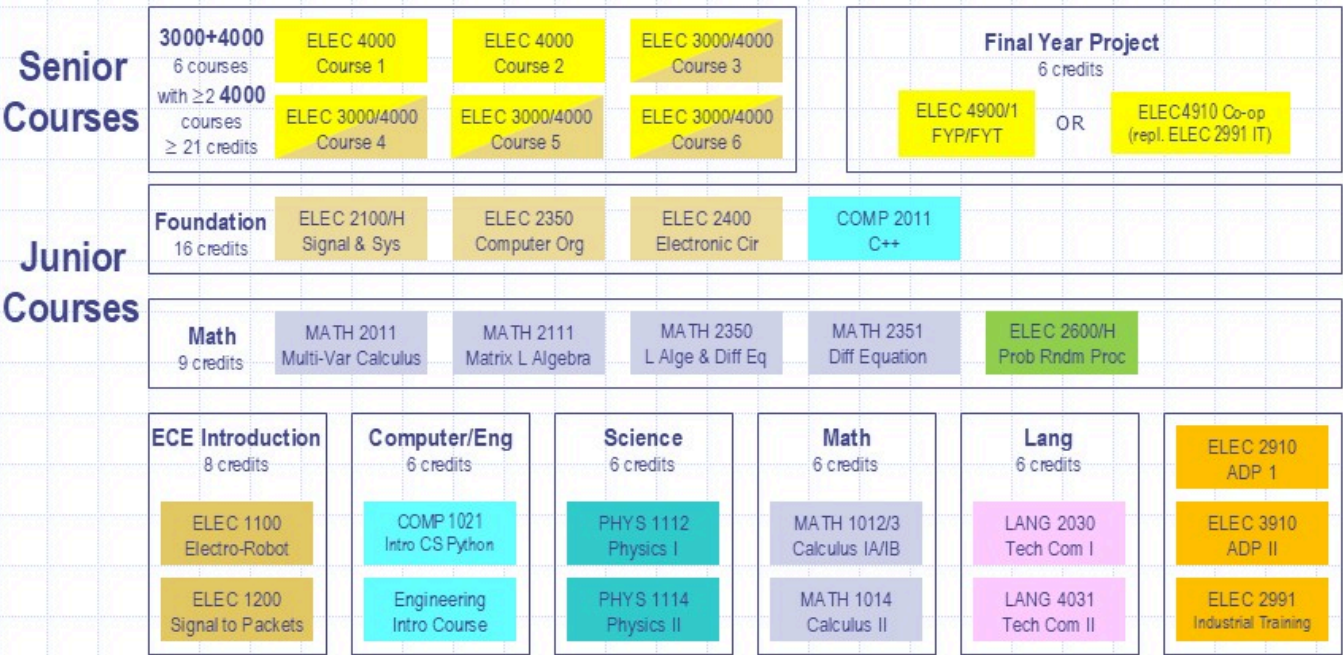
Common Core (30): Foundations (12-15); Broadening (12-18) Experiencing (0-3)

SENG Courses (24): Science (6); Math (15); Programming (3)

ECE Major (51): Introduction (8); Foundation (12); Comp (4); 3000+4000 (21); FYP/T (6)

Free Electives (15)

ELEC Program Requirement (2022/23 to 2024/25)



Graduation: 120 Credits

Common Core (30)*: Foundations (12 to 15); Broadening (12 to 18); Experiencing (0 to 3) **please refer to the University's new common core website*

SENG Prerequisites (30): Science (6); Math (15); SENG Intro (3); Programming (3); Eng (3)

ELEC Major (54): Introduction (8); Foundation (12); COMP (4); 3000+4000 (21); FYP/FYT (6); Eng (3)

Research Option (7): Engg. Research Methodology (1); Advanced Electives (6)

ELEC Program Requirement (2018/19 to 2021/22)

Senior Courses

3000+4000 6 courses with ≥ 2 4000 courses ≥ 21 credits	ELEC 4000 Course 1	ELEC 4000 Course 2	ELEC 3000/4000 Course 3	Final Year Project 6 credits ELEC 4900/1 FYP/FYT OR ELEC 4910 Co-op (repl. ELEC 2991 IT)
	ELEC 3000/4000 Course 4	ELEC 3000/4000 Course 5	ELEC 3000/4000 Course 6	

Junior Courses

Foundation 16 credits	ELEC 2100/H Signal & Sys	ELEC 2350 Computer Org	ELEC 2400 Electronic Cir	COMP 2011 C++	
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Math 9 credits	MATH 2011 Multi-Var Calculus	MATH 2111 Matrix L. Algebra	MATH 2350 L. Alge & Diff Eq	MATH 2351 Diff Equation	ELEC 2600/H Prob Rndm Proc
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ECE Introduction 8 credits ELEC 1100 Electro-Robot ELEC 1200 Signal to Packets	Computer/Eng 6 credits COMP 1021 Intro CS Python Engineering Intro Course	Science 6 credits PHYS 1112 Physics I PHYS 1114 Physics II	Math 6 credits MATH 1012/3 Calculus IA/IB MATH 1014 Calculus II	Lang 6 credits LANG 2030 Tech Com I LANG 4031 Tech Com II	ELEC 2910 ADP I ELEC 3910 ADP II ELEC 2991 Industrial Training
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Graduation: 120 Credits

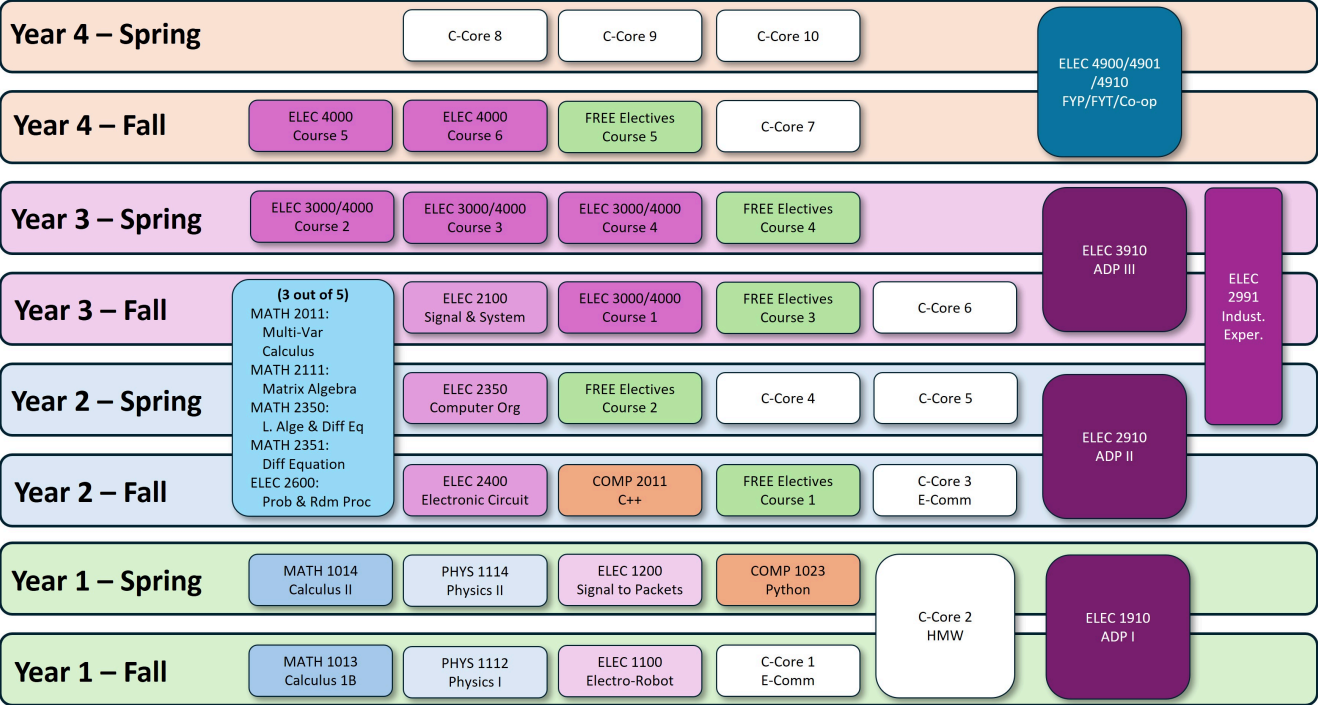
Common Core (36): Humanities (6); Social Analysis (6); Sc & Tech (6); Quantitative Reasoning (3), Eng (6); Chin (3); electives (6)

SENG Prerequisites (30): Science (6); Math (15); SENG Intro (3); Programming (3); Eng (3)

ELEC Major (54): Introduction (8); Foundation (12); COMP (4); 3000+4000 (21); FYP/FYT (6); Eng (3)

Research Option (7): Engg. Research Methodology (1); Advanced Electives (6)

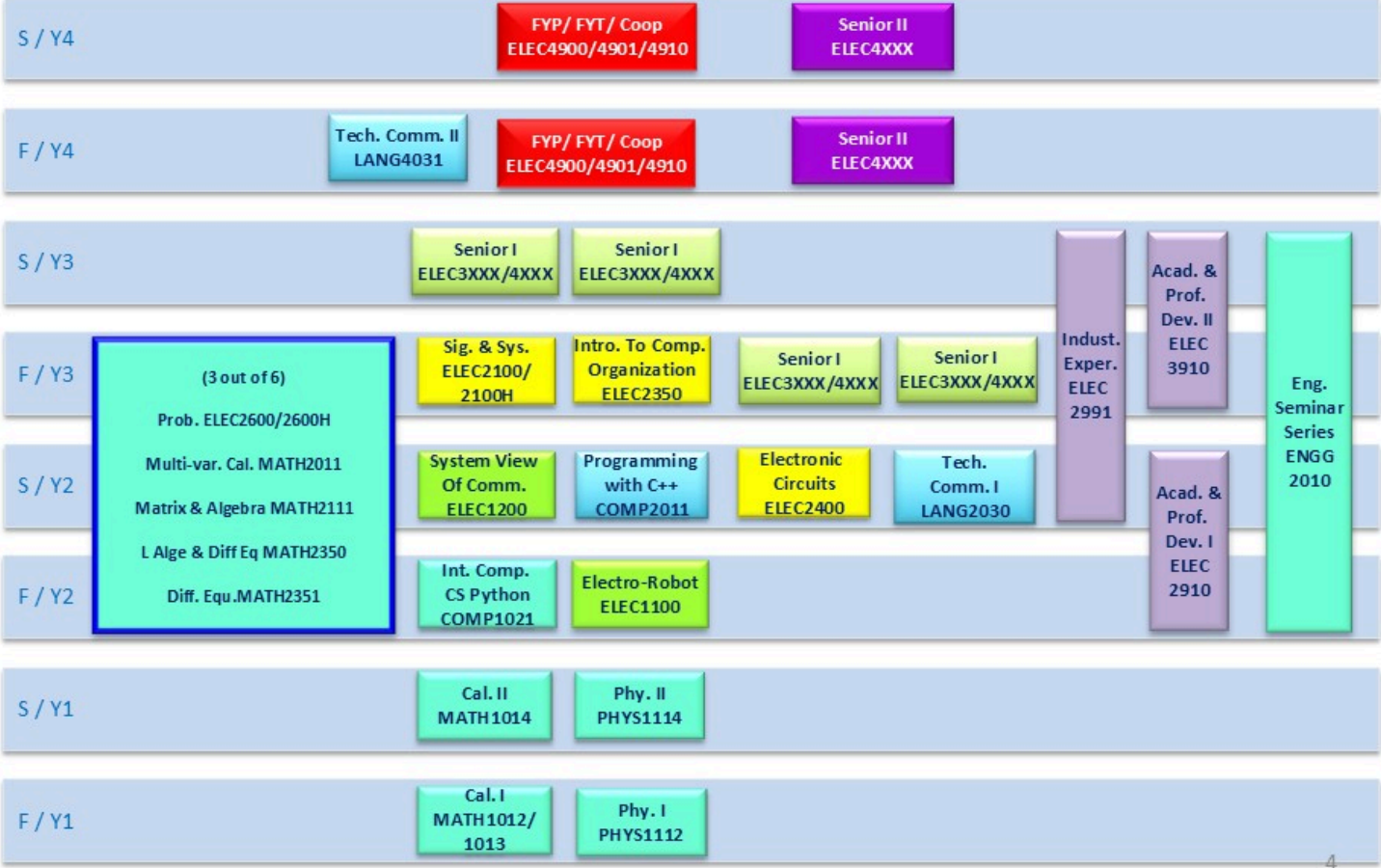
Sample ELEC Study Pattern (2025/26 or later)



Curriculum Structure –
Electronic Engineering (2022/23 to 2024/25)

May be taken in one of these terms

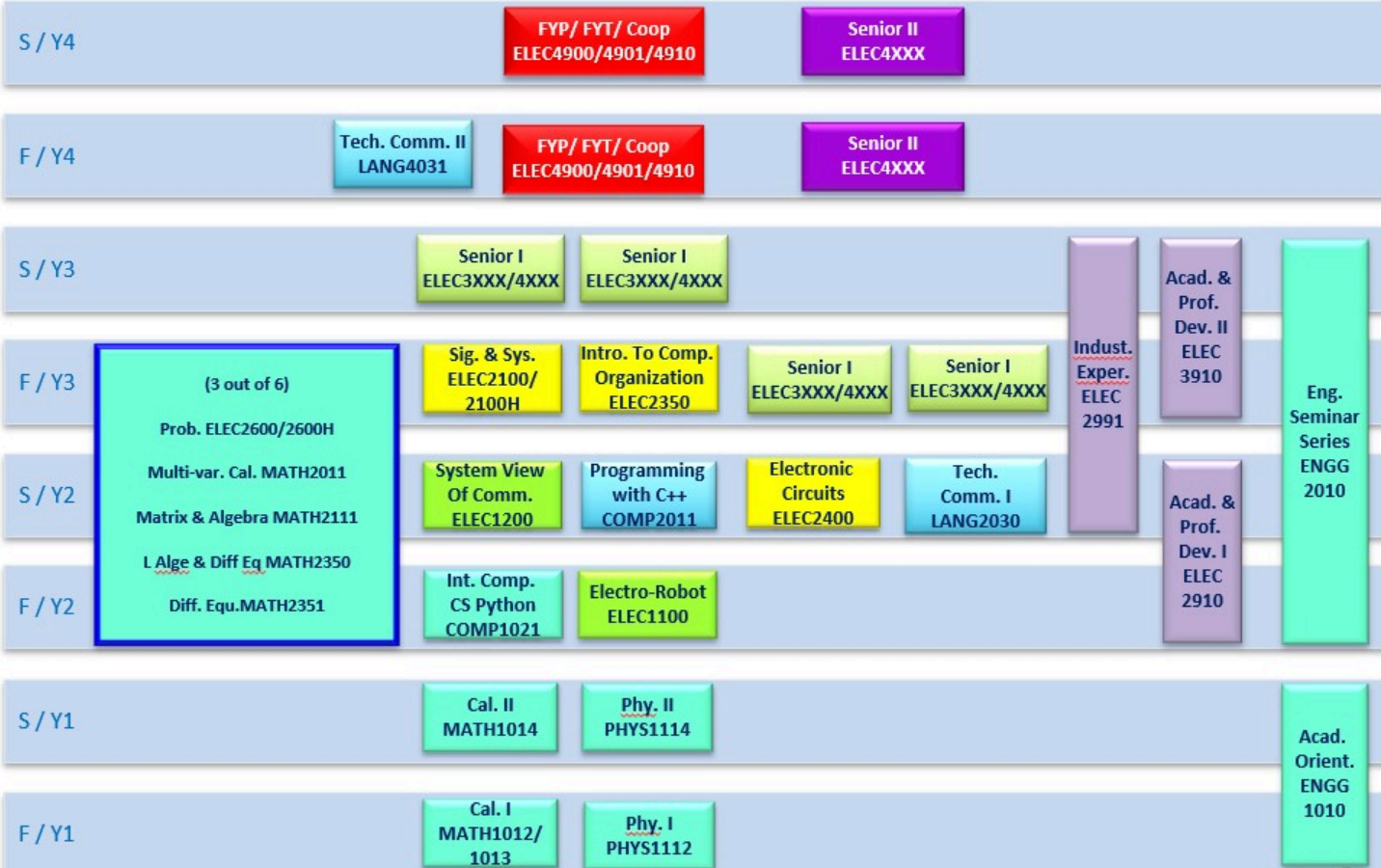
- Senior II
- Senior I
- Foundation
- ECE Intro
- Non-major



Curriculum Structure –
Electronic Engineering (2018/19 to 2021/22)

May be taken in one of these terms

- Senior II
- Senior I
- Foundation
- ECE Intro
- Non-major



Research Option

Students may opt to graduate with or without an option. Students who take the Research Option must complete all requirements specified in addition to the major requirements.

[Read More](#)

Prerequisite

Students may refer to the latest [course catalog](#) to review the prerequisite requirements of the senior courses of interest and plan their course selection accordingly.

Critical Milestones

Students should follow critical milestones to ensure that they could graduate on schedule.

2018/19 or later			^
Course code	Course name	By Term / Year	
MATH 1014	Calculus II	Fall / Year 2	
PHYS 1114	General Phys II	Fall / Year 2	
COMP 1022P	Introduction to Computing with Java	Spring / Year 2	
COMP 2011	Programming with C++	Fall / Year 3	
ELEC 1100	Intro to Electro-Robot Design	Spring / Year 2	
ELEC 1200	A System View of Communications: from Signals to Packets	Fall / Year 3	
MATH 2011 / 2111 / 2350 / 2351 AND ELEC 2600 / 2600H	Intro to Multivariable Calculus / Matrix Algebra and Applications / Intro to Differential Equations / Applied Linear Algebra and Differential Equations/ Probability and Random Processes in Engineering/ Honors Probability and Random Processes in Engineering	At least ONE by Fall / Year 3; and THREE by Spring / Year 3	
ELEC 2100 / 2100H / 2350 / 2400	Signals and Systems / Honors Signals and Systems/ Computer Organization / Electronic Circuits	ALL by Spring / Year 3	
ELEC3XXX / 4XXX		TWO by Fall / Year 4	
2017/18 or earlier			▼

University Core and Required Courses

Introductory courses show students the big picture of ECE through two lab-based courses that all students must take early in their studies. Foundation courses provide students the core knowledge of ECE through four courses that all students must take as prerequisites for higher-level courses. Science / Math courses provide essential background knowledge to prepare students for higher-level courses. All students are required to take two Science courses, two Calculus courses, and three out of four other Math courses.

University Core (2022/23 or after)	▼
University Core (2021/22 or before)	▼
Introductory / Foundation Courses	▼
Science / Math / Comp Courses	▼

Senior Courses

Senior courses broaden students' horizon and equip them with the necessary tools in different ECE areas, and allow students to deepen their knowledge within a selected area of interest. Students are required to take at least 21 credits of senior courses including at least two ELEC 4000-level courses.

Artificial Intelligence and Data Science	▼
Bioelectroincs and Bioinformation Technology	▼

Control and Robotic Systems	▼
Photonics and Microwave Technology	▼
Embedded Systems	▼
Integrated Circuits and Systems	▼
Semiconductor Devices and Technology	▼
Signal Processing and Communications	▼

Recommended Study Patterns

The recommended study patterns help students plan their studies in selected areas.

Embedded Systems + Integrated Circuits and Systems	▼
Microelectronic Devices & Technology + Electromagnetics	▼
Signal Processing and Communications + Embedded Systems	▼
System Modeling, Analysis and Control + Embedded Systems	▼

Curriculum of Other Years of Entry

Students must adhere to the curriculum of the year of entry throughout their program of study. Details of the degree and major requirements can be found in the program catalog of the year of intake.

[Read More](#)

Program Requirements

Please refer to the [program catalog](#) for the latest information

Final Year Projects/Theses

All ECE students are required to complete a Final Year Project/Thesis (FYP/T). It is an excellent opportunity for them to synthesize the theory and experimental work they learned during the Program in order to complete the project.

[Read More](#)

Industrial Experience

All ECE students are required to undertake practical training as a compulsory graduation requirement. Students could gain valuable hands-on experience through training and enhance their understanding and appreciation of the knowledge acquired in classrooms and laboratories. To satisfy the practical training requirement, students may either complete an internship or the required Industrial Training modules.

[Read More](#)

Minor Program

Undergraduate students may enroll in a minor program in addition to their disciplinary major if they meet the enrollment conditions specified. On successful completion of both major and minor programs, the student will receive a degree in their major subject with a minor in their selected program.

Minor Program in Robotics	▼
Minor Program in Sustainable Energy Engineering	▼

Students may also enroll in minor programs offered by other Schools in the University.

[Read More](#)

Double Major

With the approval of the major department, undergraduate students can graduate with dual program designation (e.g. BEng in Electronic Engineering AND Computer Engineering), if they can complete the requirements of the two undergraduate programs within the normal period of study.

[Read More](#)

Exchange Programs

The University has established exchange programs with leading institutions, such as UC Berkeley, University of Michigan, UCLA, University of Pennsylvania, Tsinghua University, University of Science and Technology in China, and is in active negotiation with a number of institutions. These exchange programs intend to provide opportunities for students to study abroad and enhance their international exposure and future career development.

[Read More](#)

Academic Advising

The ECE Department has set up several channels to help students in academic affairs: student advising courses and FYP supervisors. All students have a mentor for each year of their major study, and they should make use of these chances to ask questions and discuss academic issues (comment on courses, instructors, career planning, further study, etc.).

[Read More](#)

Professional Recognition

Our programs are recognized by the Hong Kong Institution of Engineers (HKIE). Graduates from our programs with honors will be deemed to have satisfied the educational requirement for Corporate Membership of HKIE.

Career Prospects

A wide variety of career paths are available to ECE graduates. Upon graduation, our graduates have taken positions in the industry as electronic engineers, communication engineers, system analysts and designers, information technologists, computer project managers and many other technical and managerial positions. A significant portion chooses to continue their study at the postgraduate level. Even graduates who choose not to pursue technical careers find that the engineering training they obtained at HKUST has given them a competitive advantage over their co-workers.

The list of our graduates' employers includes many world-class local and overseas companies, such as ASM, ASTRI, Fujitsu, Huawei, Hutchison, Lenovo, Microsoft, National Semiconductor, PCCW, Philips, Smartone, Solomon Systech, Varitronix, etc.

Enquiries

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